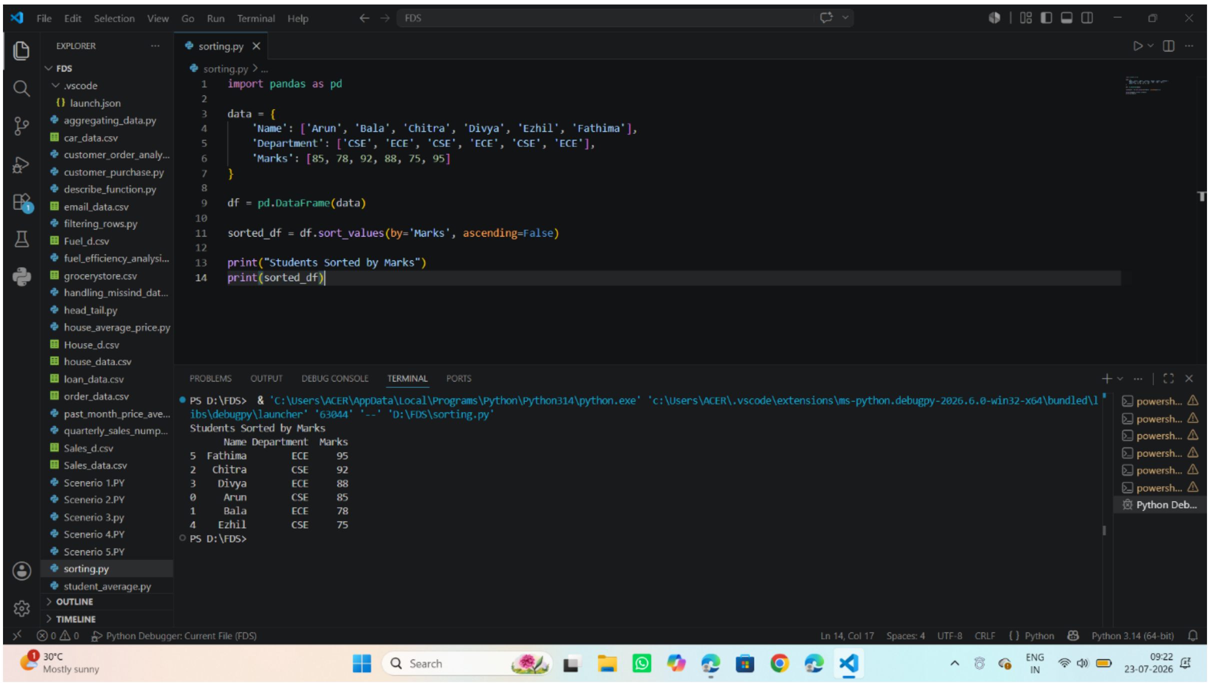
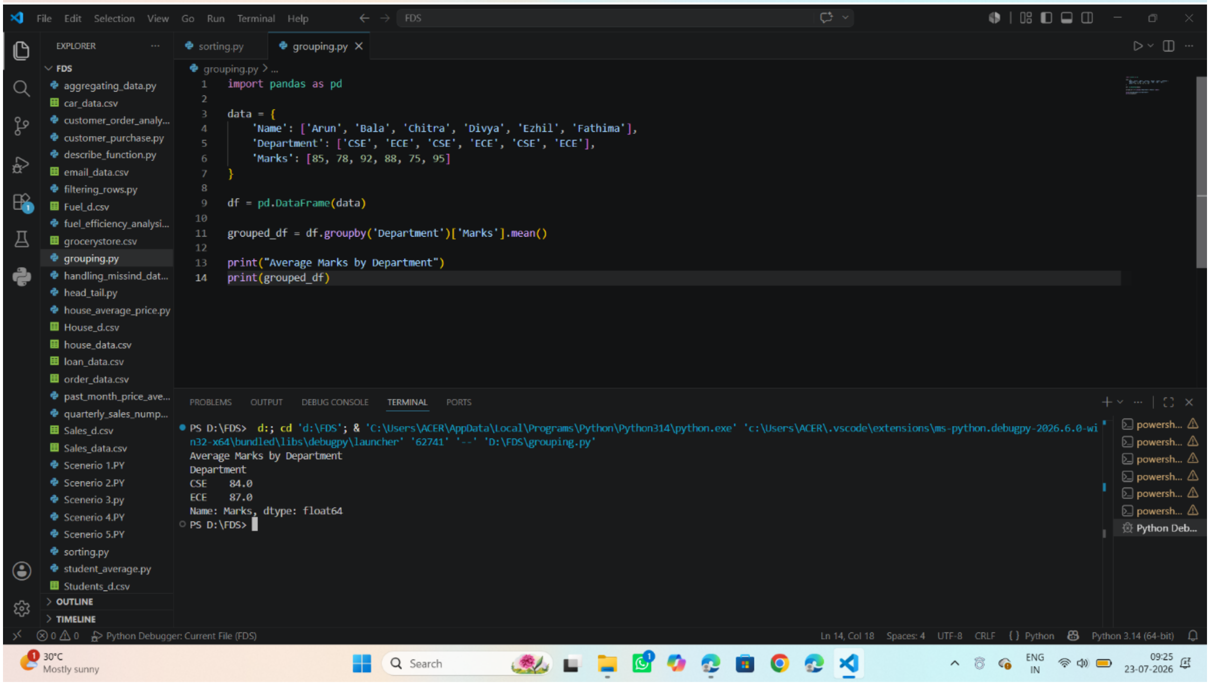
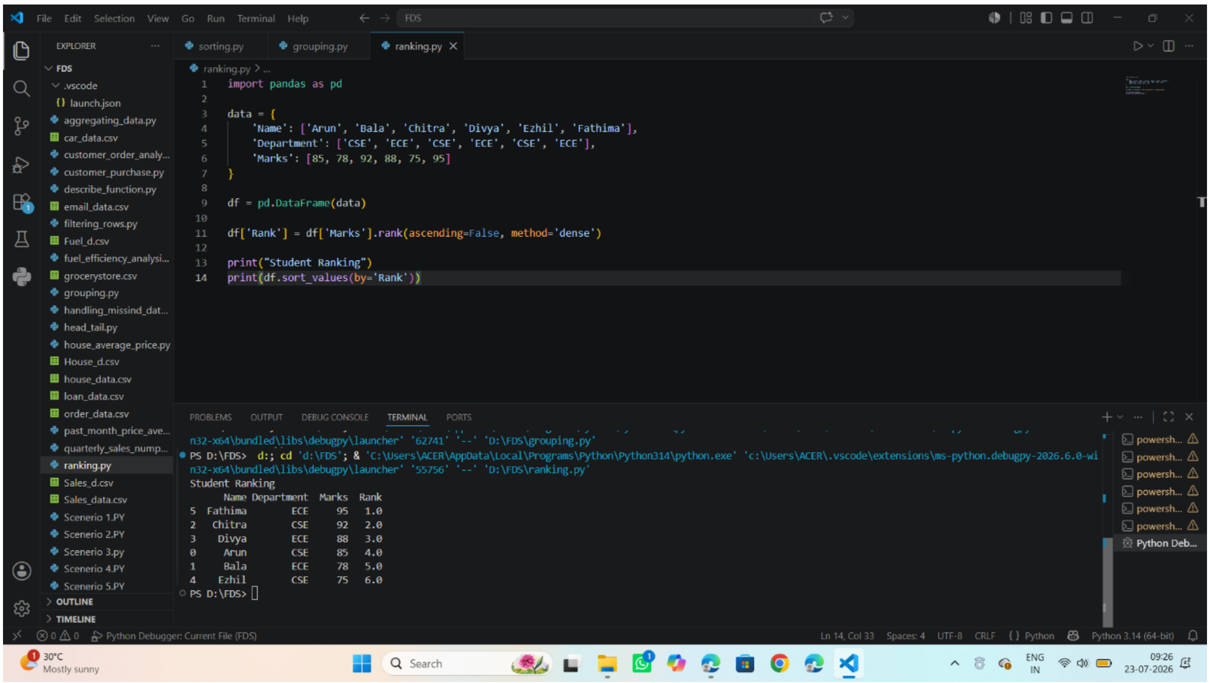

sorting.py
import pandas as pd
data = {
'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
sorted_df = df.sort_values(by='Marks', ascending=False)
print("Students Sorted by Marks")
print(sorted_df)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\FDS> & 'C:\Users\ACER\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63044' '--' 'D:\FDS\sorting.py'
Students Sorted by Marks
Name Department Marks
5 Fathima ECE 95
2 Chitra CSE 92
3 Divya ECE 88
0 Arun CSE 85
1 Bala ECE 78
4 Ezhil CSE 75
PS D:\FDS>


grouping.py
import pandas as pd
data = {
'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
grouped_df = df.groupby('Department')['Marks'].mean()
print("Average Marks by Department")
print(grouped_df)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\FDS> d:; cd 'd:\FDS'; & 'C:\Users\ACER\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62741' '--' 'D:\FDS\grouping.py'
Average Marks by Department
Department
CSE 84.0
ECE 87.0
Name: Marks, dtype: float64
PS D:\FDS>


ranking.py
import pandas as pd
data = {
'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
df['Rank'] = df['Marks'].rank(ascending=False, method='dense')
print("Student Ranking")
print(df.sort_values(by='Rank'))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Student Ranking
Name Department Marks Rank
5 Fathima ECE 95 1.0
2 Chitra CSE 92 2.0
3 Divya ECE 88 3.0
0 Arun CSE 85 4.0
1 Bala ECE 78 5.0
4 Ezhil CSE 75 6.0
PS D:\FDS>

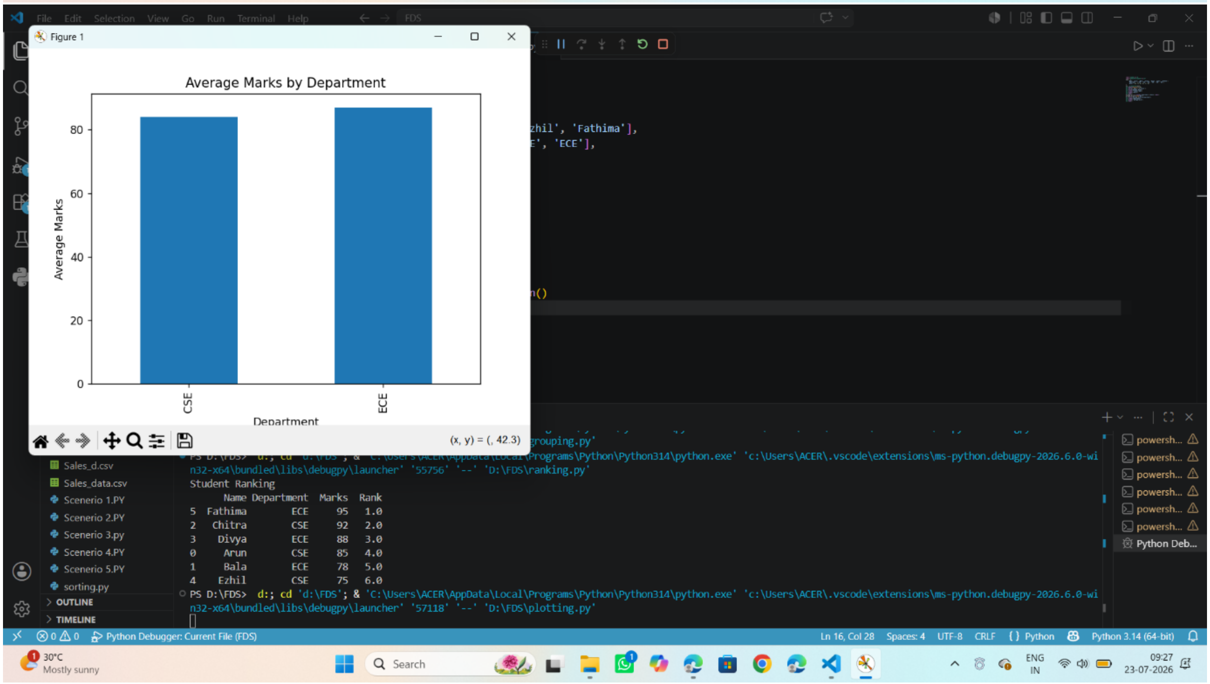


Figure 1
Average Marks by Department
Average Marks
80
60
40
20
0
CSE
ECE
Department
(x, y) = (, 42.3)
Student Ranking
Name Department Marks Rank
5 Fathima ECE 95 1.0
2 Chitra CSE 92 2.0
3 Divya ECE 88 3.0
0 Arun CSE 85 4.0
1 Bala ECE 78 5.0
4 Ezhil CSE 75 6.0

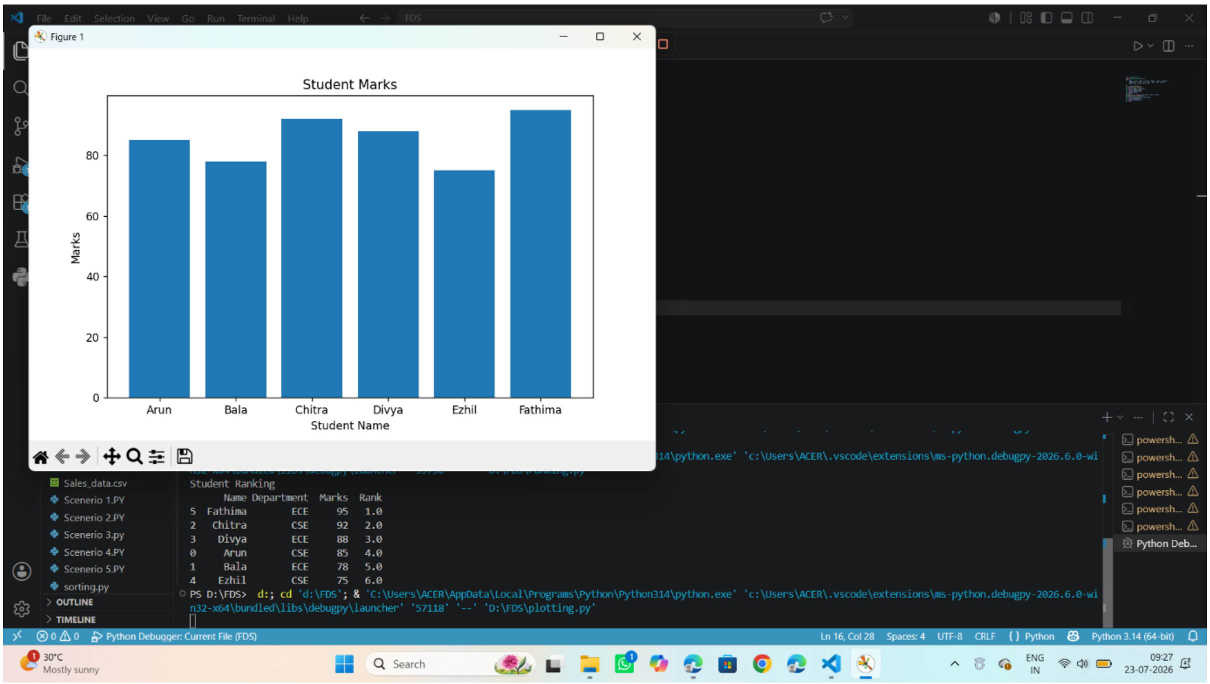


Figure 1
Student Marks
Marks
0
20
40
60
80
Arun
Bala
Chitra
Divya
Ezhil
Fathima
Student Name
Sales_data.csv
Scenerio 1.PY
Scenerio 2.PY
Scenerio 3.py
Scenerio 4.PY
Scenerio 5.PY
sorting.py
OUTLINE
TIMELINE
Student Ranking
Name Department Marks Rank
5 Fathima ECE 95 1.0
2 Chitra CSE 92 2.0
3 Divya ECE 88 3.0
0 Arun CSE 85 4.0
1 Bala ECE 78 5.0
4 Ezhil CSE 75 6.0
PS D:\FDS> d:; cd 'd:\FDS'; & 'C:\Users\ACER\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '57118' '--' 'D:\FDS\plotting.py'
Python Debugger: Current File (FDS)
Ln 16, Col 28
Spaces: 4
UTF-8
CRLF
Python
Python 3.14 (64-bit)
30°C
Mostly sunny
Search
ENG
IN
09:27
23-07-2026